

**DYNAMICAL SYSTEMS:  
CLASS SEPTEMBER 15, 2026  
6. 2D NONLINEAR SYSTEMS AND FIXED POINTS**

ROB TOMPKINS

## Teams

Post screenshots of your work to the course Google Drive today: Link to drive folder. Include words, labels, and other short notes on your whiteboard that might make those solutions useful to you or your classmates. Name your images using the format:

- teamXX-YY.EXT

where XX is your two-digit team number (e.g. 06 if you are team 6), YY is the two digit number of image for today (e.g. 01 for the first image, 02 for the second, and so on.), and EXT is the extension for the image filetype you used.

## Team Activity

**Three ways to find a conserved quantity:**

- (1) Consider a system of the form  $m\ddot{x} = F(x)$  where  $F(x) = -\frac{dV}{dx}$  with  $V(x)$  called the **potential energy**.  $E = \frac{1}{2}m\dot{x}^2 + V$  is a conserved quantity for this system.

**Example:** Let  $\ddot{x} = x^3$ .

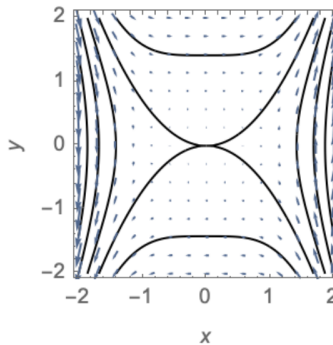
By introducing  $y = \dot{x}$ , this can be written as the system  $\dot{x} = y, \dot{y} = x^3$ . Then we have

$$V(x) = -\int x^3 dx = -x^4/4.$$

Try  $E = \frac{1}{2}y^2 - x^4/4$ . Is this conserved? (It should be).

$$\frac{dE}{dt} = y\dot{y} - x^3\dot{x} = yx^3 - x^3y = 0.$$

It is conserved!



**Problem 1:** Let  $\ddot{x} = -kx$ . Find  $E$  and check that it is conserved.

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- (2) A **phase curve** of a system  $\dot{x} = f(x, y), \dot{y} = g(x, y)$  is a curve  $y = Y(x)$  (note that  $\dot{y} = Y_x \dot{x}$ ) that obeys the differential equation  $\frac{dY}{dx} = \frac{\dot{y}}{\dot{x}} = \frac{g(x, Y)}{f(x, Y)}$ . Trajectories that start on such a curve stay on it. When  $\frac{dy}{dx} = \frac{g(x, y)}{f(x, y)}$  is a separable differential equation, the process of solving it can also yield a conserved quantity.

**Problem 2:** A model of two species with exponential population growth and competition is given by

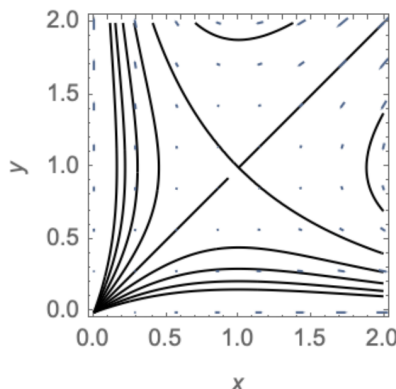
$$\begin{aligned} N_1' &= r_1 N_1 - \alpha_1 N_1 N_2 \\ N_2' &= r_2 N_2 - \alpha_2 N_1 N_2. \end{aligned}$$

The model nondimensionalizes to

$$\begin{aligned} x' &= x - xy \\ y' &= y(\rho - x). \end{aligned}$$

This nondimensionalization requires assuming that each population has its own units, so number of rabbits and number of sheep were each be assigned their own unknown constant in the nondimensionalization process.

- Using the nondimensional system, find an expression satisfied by most phase curves. Solve a separable differential equation to do this. You don't need to write it in the form  $y = Y(x)$ ; an implicit relationship between  $x$  and  $y$  is fine.
- Argue that the quantity  $H(x, y) = y - \ln y + \rho \ln x - x$  is almost always conserved. Which trajectories don't work with this function?



- (3) A **Hamiltonian function**,  $H(x, y)$  is an energy function. The associated dynamical system  $\dot{x} = \frac{\partial H}{\partial y}, \dot{y} = -\frac{\partial H}{\partial x}$  is called a **Hamiltonian system**. In a Hamiltonian system, the Hamiltonian is a conserved quantity.

**Problem 3:** Let  $\dot{x} = x^2, \dot{y} = -2xy + x$ . Show this system is Hamiltonian.

**Problem 4:** Consider the system

$$\begin{aligned} \dot{x} &= -\mu y + xy \\ \dot{y} &= \mu x + \frac{1}{2}(x^2 - y^2). \end{aligned}$$

Assume  $\mu > 0$ .

- (a) This system has a conserved quantity. Assume  $\dot{x} = H_y$  and  $\dot{y} = -H_x$  for an unknown function  $H(x, y)$ . If our assumption is correct then  $\partial_x \dot{x} = -\partial_y \dot{y}$ . Why is that?

- (b) Show that  $\partial_x \dot{x} = -\partial_y \dot{y}$  for this system.
- (c) Construct a function  $H(x, y)$  that is conserved for this system.
- (d) If time permits, begin the process of constructing a phase portrait for this system.

### REVERSIBLE SYSTEMS:

A system of the form

$$\dot{x} = f(x, y)$$

$$\dot{y} = g(x, y)$$

is **reversible** if the change of variables  $t \mapsto -t$  and  $(x, y) \mapsto R(x, y)$  with  $R^2(x, y) = (x, y)$  yields the same system, i.e. the system is invariant to this change of variables.

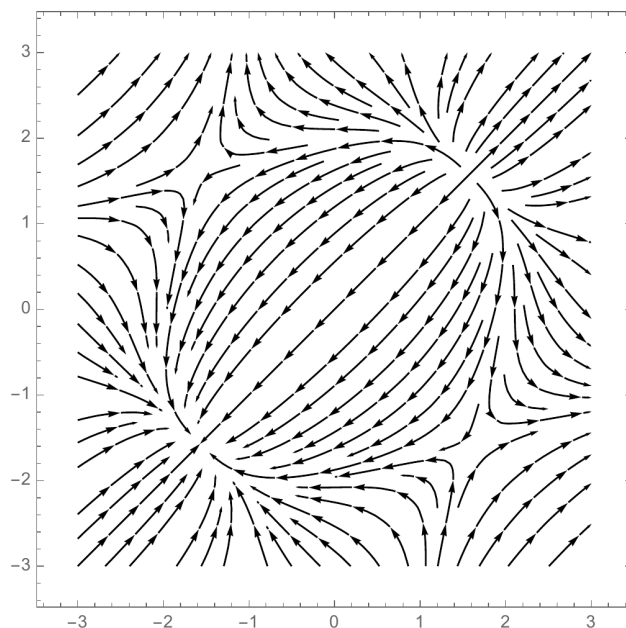
- examples of  $R$ :  $R(x, y) = (x, -y)$  (mechanical systems,  $m\ddot{x} = F(x)$ ) or  $R(x, y) = (-x, -y)$ . In 2D, a reflection about any axis through the origin has this property.
- e.g. if  $f$  is odd in  $y$  ( $f(x, -y) = -f(x, y)$ ) and  $g$  is even in  $y$  ( $g(x, -y) = g(x, y)$ )
- Reversible systems are “ideal” in the sense that no energy is “wasted.”
- Real systems are characterized by friction, turbulence, unrestrained expansion, temperature gradients and mixing of dissimilar substances and are therefore irreversible.
- Similar to conservative systems, but not the same thing, e.g.

$$\dot{x} = y, \quad \dot{y} = x^3$$

is both conservative and reversible but

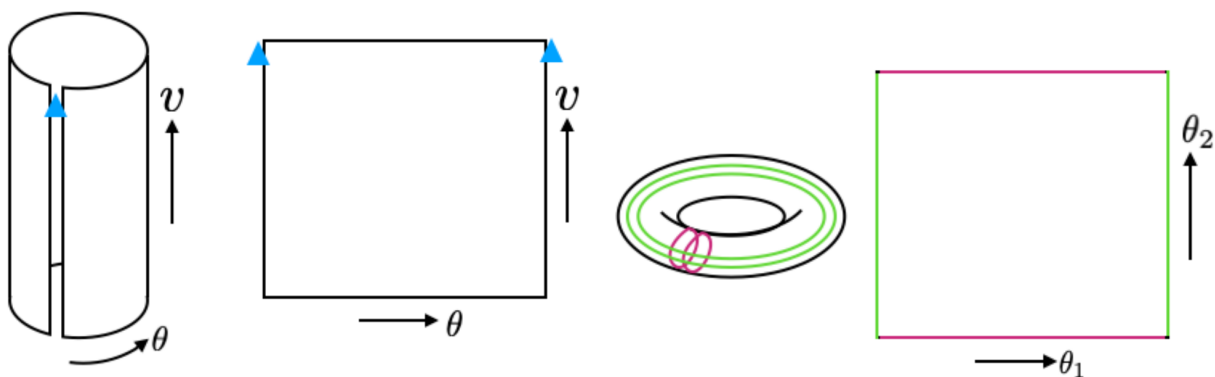
$$\dot{x} = -2 \cos x - \cos y, \quad \dot{y} = -2 \cos y - \cos x$$

is only reversible. (How can you tell it is not conservative from the phase portrait below?)



A **cylindrical phase space** arises when one coordinate can take on any value in  $\mathbb{R}$  (the real numbers) while the other coordinate is an angle.

A **toroidal phase space** arises when two coordinates are angles.



**Problem 5:** To get used to a phase space that is a torus, think about two oscillators that are not interacting (like the hour hand and the minute hand on a clock: they each go at their own pace, and that pace is constant).

- Let  $\dot{\theta}_1 = 1$  and  $\dot{\theta}_2 = 2$ . If the oscillators each start at a phase angle of zero, so at the point  $(0, 0)$ , draw their trajectory onto the phase space. Use a square to represent the space. Will the pair of oscillators pass through  $(0, 0)$  at some point?
- Now let  $\dot{\theta}_1 = \pi$  and  $\dot{\theta}_2 = 2\pi$ . With an initial condition of  $(0, 0)$ , draw their trajectory onto the phase space. How is the trajectory different from the one in part a?
- Let  $\dot{\theta}_1 = \pi$  and  $\dot{\theta}_2 = \sqrt{2}\pi$ . Assume the oscillator pair again starts at  $(0, 0)$ . The first oscillator will return to a phase of zero at time 2, time 4, etc. When does the second oscillator return to a phase of zero? Will the pair pass through  $(0, 0)$  at some point?

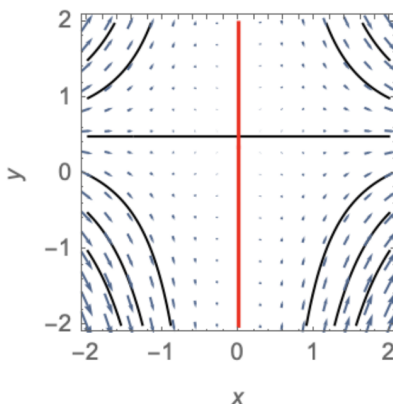
## ANSWERS

1) Potential energy practice:  $\dot{x} = y, \dot{y} = -kx$ .  $V(x) = -\int -kx dx = kx^2/2$ . Try  $E = \frac{1}{2}y^2 + kx^2/2$ .  $\dot{E} = y\dot{y} + kx\dot{x} = y(-kx) + kxy = 0$ .

3) Hamiltonian example: Assume  $H_y = x^2$ , so  $H = x^2y + g(x)$  where  $g(x)$  is unconstrained by  $H_y$ .

Using this  $H$ , compute  $H_x$  to find  $H_x = 2xy + g'(x)$ . We want  $-\dot{y} = H_x$ , so  $-2xy + x = -2xy - g'(x)$ . This implies that  $g'(x) = -x$ , so  $g(x) = -x^2/2 + c$ . We can choose  $c = 0$  for simplicity.

So we have  $H(x, y) = x^2y - x^2/2$  as a conserved quantity.

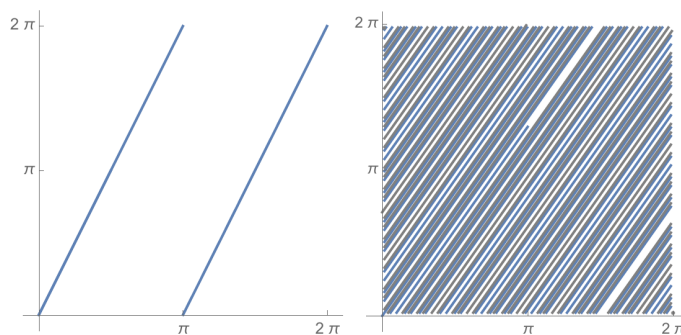


4) Hamiltonian practice: a: Mixed partials are equal,  $H_{xy} = H_{yx}$ , so if  $\dot{x}$  and  $\dot{y}$  are given by those partials, then we can use  $\dot{x}$  and  $\dot{y}$  to compute the mixed partials in two different ways.

b:  $\partial_x \dot{x} = y$ .  $-\partial_y \dot{y} = -\frac{1}{2}(-2y) = y$ . These are equal.

c:  $H_y = -\mu y + xy$  so  $H(x, y) = \frac{1}{2}y^2(-\mu + x) + G(x)$ .  $-H_x = -\frac{1}{2}y^2 - G_x = \mu x + \frac{1}{2}(x^2 - y^2)$  so  $G_x = -\mu x - \frac{1}{2}x^2$ .  $G(x) = -\mu \frac{1}{2}x^2 - \frac{1}{6}x^3 + c$ . I just need a single function that works, not a whole family of them.  $H(x, y) = -\mu \frac{1}{2}x^2 - \frac{1}{6}x^3 + \frac{1}{2}y^2(-\mu + x)$  is one such function.

5a: (see below left)  $\theta_1$  moves 1 unit in the time  $\theta_2$  moves two units, so the trajectory is a line with slope 2 through  $(0, 0)$ . It will leave at the top (at  $(\pi, 2\pi)$ ) and come back in at the bottom (at  $(\pi, 0)$ ).



5b: This trajectory will look the same as above. We're just moving faster along the same path.

5c: (see above right) oscillator 1 returns to phase 0 at time 2, 4, etc, so times  $2\mathbb{Z}^+$ . oscillator 2 returns to a phase of 0 when  $\sqrt{2}\pi t = 2n\pi$  for  $n$  an integer  $\Rightarrow \sqrt{2}t = 2n$  or when  $t = \sqrt{2}n$ . The return times are not integers. The pair of oscillators will not both return to zero at the same time, so this trajectory does not pass through  $(0, 0)$  again!